

Program : Diploma in Architecture	
Course Code : 6189	Course Title: Estimating and Costing Lab
Semester : 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3(L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To learn the procedure for estimating and costing of Civil Engineering works.
- To perform rate analysis for different items associated with construction projects.
- To use software for detailed estimate related to civil infrastructural projects

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Engineering Mathematics	1
Basic knowledge about building materials and construction methods		Construction materials and engineering	3
Basic knowledge about building Drawing		Architectural drawing, I	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand Basic concepts, techniques and application of estimation and costing	12	Applying
CO2	Prepare detailed estimate from given drawing	12	Applying
CO3	Analyse quantities in finishing work	10	Applying
CO4	Analyse rates and abstract of estimate	9	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						3
CO2	2						3
CO3	2						3
CO4	2						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand Basic concepts, techniques and application of estimation and costing		
M1.01	Define the terms quantity surveyor, quantity surveying, and understand elements of estimate, different types of estimate	1	Understanding
M1.02	List out duties of quantity surveyor,	1	Understanding
M1.03	Find approximate cost of the building using plinth area method, cubic content method and service unit method	3	Applying
M1.04	Identify different units of measurements and accuracy of measurements.	1	Understanding
M1.05	Define the terms sundries, Lump sum, Lead and lift, contingencies, unforeseen items, work charged establishment. Estimate PAC	1	Understanding
M1.06	Identify the different methods of earth work volume computation - Trapezoidal – midsection, mean section (trapezoidal and Prismoidal formula)	2	Understanding
M1.07	Compute the quantities of earth work from Longitudinal section and Cross section incutting and filling (no transverse slope)	2	Applying
M1.08	Compute Capacity of reservoir from contour map using trapezoidal rule and prismoidal	1	Applying
CO2	Prepare detailed estimate from given drawing		
M2.01	Describe different methods of taking quantities by centre line method, long and short wall method	1	Understanding

M2.02	Compute the estimate of,doors, windows and ventilators, office building with RCC roof	3	Applying
M2.03	Compute the quantities for One roomed building (RCC roof- Flat & Sloped), Two roomed building (RCC roof-Flat & Sloped) except finishing work	3	Applying
M2.04	Compute the quantities for residential building with RCC roof-(Flat & Sloped) ,office building with RCC roof Flat & Sloped), Two roomed building (RCC roof-Flat & Sloped) except finishing work	5	Applying
	Series Test I	1	
CO3	Analyse quantities in finishing work		
M3.01	Compute quantities of all finishing work,water supply and sanitary items for One roomed building (RCC roof-Flat & Sloped), Two roomed building (RCC roof-Flat & Sloped)	4	Applying
M3.02	Compute quantities of all finishing work, water supply and sanitary items for residential building with RCC roof-Flat & Sloped, office building with RCC roof.	3	Applying
M3.03	Compute the quantities of masonry well, RCC water tank and WBM road	3	Applying
CO4	Analyse rates and abstract of estimate		
M4.01	Define cost of materials at source and at site – conveyance charges – Lump sum items -extra labour – contractor's profit	3	Understanding
M4.02	Identify standard data book -schedule of rates Prepare conveyance statement	3	Understanding
M4.03	Prepare rate analysis using standard data book, conveyance statement and schedule of rate	3	Applying
	Series Test II	1	

Text /Reference:

T/R	Book Title/Author
T1	Datta, B.N., Estimating and Costing in Civil engineering, UBS Publishers Distributors
R2	Peurifoy, Robert L. Oberlender, Garold, Estimating construction cost (fifth edition), McGraw Hill Education, ,New Delhi.
R3	Rangwala, S.C., Estimating and Costing, Charotar Publishing House, Anand.
R4	Birdie, G.S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd. Delhi
R5	Patil, B.S., Civil Engineering Contracts and Estimates, Orient Longman, Mumbai

Online Resources:

Sl. No	Website Link
1	https://www.iare.ac.in/sites/default/files/lecture_notes/IARE_E%26C_Lecture_NOTES.pdf
2	http://www.alphace.ac.in/downloads/notes/cv/10cv73.pdf www.alphace.ac.in › downloads › notes
3	http://fmcet.in/CIVIL/CE6704_uw.pdf
4	https://easyengineering.net/estimating-costing-specification-valuation-in-civil-engineering-by-m-chakraborti/
5	https://www.academia.edu/29396015/Estimating_Costing_in_Civil_Engineering_Basic